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Capability of multimodal large language models to describe gravitational wave images

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Abstract

The advent of multi-messenger astronomy has highlighted the necessity for a comprehensive analysis of immense datasets generated by multiple astronomical observation instruments. The field of gravitational wave astronomy has employed deep learning for the efficient processing of data. However, the utilization of this technique has predominantly been confined to a single type of input data, such as images. While image-to-text applications with multimodal large language models (LLMs) are growing in other fields, a notable gap remains in the generation of textual descriptions of astrophysical and astronomical images. This study explores the potential of using multimodal LLMs to describe spectrogram images of anomalies captured by ground-based gravitational wave observatories. Three models, GPT-4, GPT-4o, and Gemini 1.5 Pro, were provided only with an image and a single prompt, without additional context. The generated text responses were then evaluated for length, meaning, and accuracy of the answers. All three models correctly identified each image as a spectrogram representation of gravitational wave data and generated appropriate explanations. However, they exhibited difficulties in detailed signal descriptions and source inference under context-free setting, indicating that further research is needed before practical application to gravitational wave data analysis. This study demonstrates the feasibility of using multimodal LLMs to transform spectrogram images into textual data and provides insight into their potential integration into gravitational wave data analysis workflows.

1. Introduction

In contemporary astronomy and astrophysics, the advent of the multi-messenger astronomy era [1] has driven a growing demand for the efficient processing of large volumes of multimodal astronomical data [2]. This highlights the necessity of integrating and analyzing multiple types of data obtained through different observational techniques to study a single astronomical phenomenon [3]. As the field that ushered in the current multi-messenger era, gravitational wave astronomy has increasingly incorporated machine learning into its data processing pipeline for tasks such as data quality analysis, waveform modeling, signal detection, and parameter estimation [4, 5]. Specifically, deep learning techniques are employed in the analysis of transient events captured by ground-based gravitational-wave detectors to ensure the quality of the data. Deep learning techniques such as convolutional neural networks (CNNs) have been widely used to analyze and classify spectrogram images of transient noise artifacts. These artifacts mimic genuine gravitational wave signals, thereby degrading detector sensitivity and necessitating advanced machine learning methods for effective data-quality management [6, 7].

However, current deep learning applications in gravitational wave astronomy face challenges not only in efficiently handling the increasing amount of real-time data but also in their predominant focus

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on single-modality data analysis. The reliance on a singular data modality, such as images, limits the resilience of deep learning models and their ability to exploit complementary multimodal information. Additionally, while it has been demonstrated that multimodal training incorporating textual descriptions of images outperforms single-modality approaches [8], such utilization of multimodal datasets remains unexplored in gravitational wave data analysis. Therefore, there is a necessity to develop multimodal gravitational wave datasets and adopt a multimodal deep learning approach to overcome the limitations of previous methods and enhance the accuracy of the deep learning models. Shifting towards multimodal data analysis would meet the inherent needs of multi-messenger astronomy, which requires integrated analysis of diverse signals from multiple instruments [9], enabling a more comprehensive interpretation of astrophysical events emitting gravitational waves [10].

Recently, large language models (LLMs) optimized for rapid and efficient data processing and multimodal fusion have been increasingly adopted in various scientific research fields. LLMs overcome the common limitations of traditional deep learning, including constraints on parallel processing, scalability issues, and overfitting [11, 12]. The advent of multimodal LLMs, such as GPT-4 V, has led to a significant paradigm shift in various fields that require the integrated analysis of multimodal data. Notably, research on generating text descriptions from image data using multimodal LLMs has been actively pursued in areas such as image captioning and the interpretation of medical and scientific images [13–15]. However, their application in the analysis of astronomical and astrophysical images remains largely unexplored.

To address this gap, we explored the capability of multimodal LLMs for image-to-text conversion. We use time-frequency spectrograms obtained from ground-based gravitational wave detectors to generate textual descriptions. The objective of this study is to generate and analyze descriptive text from gravitational wave image data using multimodal LLMs. The contributions of this study are as follows: (1) this study conducts a preliminary examination of whether multimodal LLMs can recognize gravitational wave spectrograms and generate descriptive text without context information; (2) it analyzes the content and meaning of text response generated from the input image; and (3) it assesses the quality of LLM-generated descriptions of gravitational wave detector signals and reports the findings. This image-to-text generation facilitates the development of multimodal datasets integrating gravitational wave images with corresponding textual information. This marks an initial step toward the transition from conventional single-modality data analysis to multimodal approaches. Furthermore, it demonstrates the feasibility of integrating multimodal LLMs into gravitational wave data analysis pipelines for practical applications.

2. Related work

In this section, we provide a brief overview of previous studies that have generated text from images across various domains. In addition to reviewing these studies, we present a gap in the existing research, identify potential research areas, and explain the necessity of further research to apply LLMs to more specialized scientific fields, such as gravitational wave astronomy.

2.1. Text generation from astronomical and astrophysical data images

In the fields of astronomy and astrophysics, textual representations of images typically rely on human authors to describe the observed features. However, the necessity of analyzing and categorizing a vast number of images has resulted in the implementation of crowdsourcing, which involves enlisting the assistance of citizen volunteers for tasks such as image classification and feature finding [16]. In particular, regarding the classification of radio galaxy images, citizen scientists not only classified the images but also contributed to their analysis through an online forum. This successful approach was also adopted for classifying transient noise artifacts in the gravitational wave data analysis process. Bowles *et al* [17] employed a citizen scientist engagement platform to create a dataset of English descriptions of radio galaxy images, whereas Chen *et al* [18] utilized text data from the Galaxy Zoo forum for multimodal classification. The Gravity Spy project by Zevin *et al* [19] integrated citizen science and machine learning to classify spectrogram images of glitches, which are transient noise artifacts from the laser interferometer gravitational wave observatory (LIGO) detector. Citizen scientists contributed to the identification and discussion of new glitch classes by proposing descriptions of their morphological features [20].

Some studies have attempted to annotate astrophysical images using computational methods. In a previous study by Bowles *et al* [21], a language model pre-trained on radio galaxy classification with embedded tokens and a random forest model was used to generate English annotations. Additionally, Fluke *et al* [22] used the sports performance analysis software, SPORTSCODE, to examine an external

galaxy's neutral hydrogen spectrum data cube. This study demonstrated that the software tool is capable of supporting the visual discovery process, including annotations.

2.2. Image-to-text generation using LLMs

The advent of deep learning and language models has led to the adoption of LLMs across various fields, with text generation from images being a particularly notable application. Prior studies have employed LLMs to generate and evaluate text that describes images, particularly in domains such as image captioning and medical diagnostics.

Initially, LLMs were integrated with multimodal models, such as bootstrapping language-image pre-training (BLIP) and contrastive language-image pre-training (CLIP). For example, CLIP and GPT-2 were integrated by Tewel *et al* [23] and Fei *et al* [24] to create an image captioning model. Gupta and Shete [13] developed a method for captioning fashion images by combining the BLIP with open pre-trained transformer (OPT) models. Ji *et al* [25] developed the ClinicalBLIP model using the InstructBLIP model as the baseline and Flan-T5-XL as the LLM to generate medical reports for medical images such as x-rays. Zhong *et al* [26] developed a model that extracts regional information and features from chest x-ray images using a multiscale region feature encoding approach. The bidirectional encoder representations from transformers model was used to produce radiology reports and survival predictions. Additionally, Chen *et al* [27] presented a pipeline that combined BLIP and Llama 2 for the automated interpretation of fundus fluorescein angiography images to support report generation and answer questions.

Alternatively, CNNs have been employed as visual encoders, and LLMs have been utilized to generate more sophisticated text. Danu *et al* [28] combined CNN and LLM to enhance the medical diagnosis process. The images were initially processed using a CNN and the resulting data was fed into the RadBloomz model to generate diagnostic findings. The diagnosis was evaluated by comparison with the actual diagnosis. Similarly, Wu *et al* [29] conducted a study in which a CNN and human experts extracted features from medical images and made diagnoses by inputting these features into ChatGPT 3.5, ChatGPT 4.0, and Bard for comparison. Goudar *et al* [30] used a CNN for image feature extraction, and the outputs were subsequently fed into GPT-3 to generate multilingual captions. Qing *et al* [31] introduced LLMs into the domain of agricultural diagnosis by employing the YOLO network to process agricultural images and GPT-4 to generate diagnostic reports.

As multimodal LLMs have evolved, studies have emerged that evaluate interpretive texts generated by directly inputting images into these models. This approach has been actively applied in the medical field for the generation of diagnoses, where MiniGPT-4 was employed for radiology report generation [32], and GPT-4 Vision (GPT-4 V) was used for radiological image analysis [33] to evaluate the diagnostic abilities of multimodal LLMs in radiological images. Khan *et al* [34], Reith *et al* [14], Suh *et al* [35], and Günay *et al* [36] recently conducted a comparative study of medical image analysis. The capabilities of various multimodal models including GPT-4 V and Gemini were also examined. In the field of biology, Yamagata *et al* [37] assessed the capacity of LLMs to annotate images derived from biological review articles using GPT-4 and GPT Turbo. Additionally, Wang *et al* [15] and Hindy *et al* [38] employed multimodal LLMs to interpret complex biological figures and bacterial images. Multimodal LLMs are also used in educational contexts. For instance, Alasadi and Baiz [39] examined GPT's explanations of handwritten chemistry problem images and discussed their educational implications. Lee *et al* [40] evaluated student-drawn images of scientific models using Gemini Pro and GPT-4 V.

2.3. Text generation from astronomical and astrophysical images using LLMs

The application of LLMs in the analysis of astronomical and astrophysical images is a relatively recent advancement and remains an under-explored area of research compared to other fields. Smirnov [41] conducted experiments using GPT-4 V, in which images of asteroid resonance angles were inputted to classify them under three labels. The findings of this study indicate that LLMs are capable of achieving high levels of accuracy in the classification of astronomical images without the need for training or additional coding.

2.4. Limitations of previous studies

In previous research, a gap has been identified in studies reporting the use of LLMs to process astronomical and astrophysical image data for generating descriptive and analytical texts. Despite the active application of deep learning technologies in gravitational wave data analysis and multi-messenger astronomy, the potential for integrating LLMs and multimodal data in these fields remains largely unexplored. Although research on the ability of multimodal LLMs to interpret a range of scientific images is growing, there are currently no studies that have used gravitational wave data as input into multimodal LLMs.

Furthermore, while a few prior studies have explored the ability of multimodal LLMs to interpret scientific figures and graphs, none have tested them on spectrograms, which are the primary visualization method for gravitational wave signals. To the best of our knowledge, this study is one of the earliest efforts to apply LLMs in the domain of gravitational wave astronomy. We input image-type data from a gravitational wave detector into LLMs to evaluate their ability to generate descriptive text. Moreover, we conduct a comparative and analytical examination of the characteristics of the text generated by different LLMs.

3. Methods

In this study, we employ LLMs to generate descriptive text from image data in the field of gravitational wave science. Figure 1 illustrates the methodology of the study. The methodology is outlined as follows: First, the input image dataset (see figure 1(a)) was collected. The input dataset consists of images obtained from the observation running of gravitational wave detector. Subsequently, one of these images was input into the multimodal LLMs along with a prompt (see figure 1(b)) to generate a response. The text generated in response to each prompt (see figure 1(c)) was then extracted. To assess the quality of the text, the responses generated by each model were recorded and analyzed sentence-by-sentence using a text analysis table (see figure 1(d)).

3.1. Image collection

To collect image data used for gravitational wave data analysis, we reviewed the input data from previous studies that applied deep learning techniques to gravitational wave data processing. We found that the most common image representation of gravitational wave data is a spectrogram of a transient event. A spectrogram is a visual representation of the frequency spectrum of a signal over time. It is widely used in various scientific fields, including physics and astronomy, to analyze the frequency components of complex signals. We analyzed datasets from studies that used spectrogram images as input for the development of deep learning models. From these, an image dataset that met the following criteria was selected: (1) it is publicly available; (2) no additional processing is required; (3) there are clear distinctions between each image within the dataset; and (4) the morphological features of the images can be readily described. The input dataset consists of glitch spectrogram images, which represent transient events from ground-based gravitational wave detectors and are publicly available online. The requisite number of image samples was determined, and images were collected from open data sources published online. For transparency and reproducibility of the experiment, no modifications or edits were made to the original images.

3.2. Text generation

A comprehensive list of LLMs utilized in prior image-to-text generation studies was compiled through a comparative analysis of the features and capabilities of each model. Based on this list, we selected multimodal LLMs that could process images as input. Among the multimodal LLMs capable of text generation from images, we employed GPT-4 and GPT-4o from OpenAI's GPT and Gemini 1.5 Pro from Google's Gemini, via their respective web interfaces. They supports images as inputs and can perform a wide range of multi-modal tasks, including image and file analysis. The experiment was conducted in August 2024. At the time of the experiment, GPT-4o was the latest publicly available version of ChatGPT, whereas GPT-4 served as a legacy model for comparison. Gemini 1.5 Pro from Google's Gemini series was also included. Including these three models aimed to ensure the reliability of the results and enable two types of comparisons: (1) GPT-4 vs GPT-4o to examine differences between successive versions on the same platform, and (2) GPT-4o vs Gemini 1.5 Pro to assess differences between contemporaneous models from different platforms. Both models were released in May 2024 and were expected to reveal differences in their respective characteristics.

While spectrograms are a general-purpose signal-processing visualization, they are also widely adopted representations in gravitational wave data analysis. Their two-dimensional morphological features play a central role in glitch identification and classification. To the best of our knowledge, although prior studies have examined LLMs' ability to interpret scientific images, no work has specifically employed spectrograms as input. For this reason, we deliberately focused on testing whether LLMs could recognize such visualization type in the absence of any contextual information. The experiment was conducted with a single, concise prompt, 'Describe the image,' and no tests were run with alternative prompts. This minimal prompt, combined with a one-shot setting, removes contextual cues and

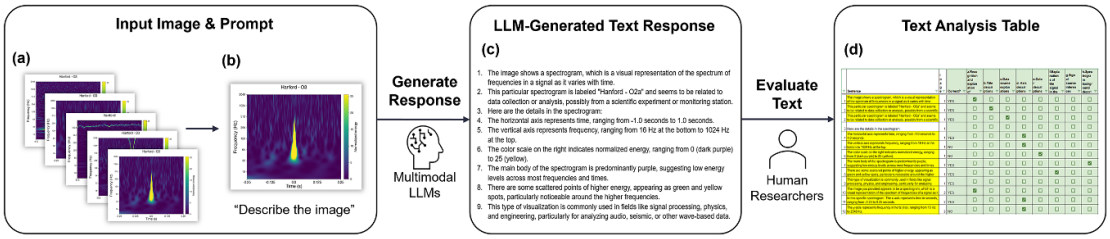


Figure 1. Overview of the methodology employed in the present study. (a) The input image dataset was collected. (b) One of these images was input along with a prompt into the multimodal LLMs. (c) The LLM-generated text response was then extracted and recorded. (d) The text analysis table is generated to analyze and evaluate these responses.

prior information about the input image, making it particularly suitable for establishing a baseline evaluation. In previous research on LLMs for image-to-text generation, similarly minimal prompts such as 'An image of' [42], 'Describe the image' [43], and 'Describe this image in detail' [44] were frequently employed as initial prompt settings to assess baseline performance. The text response was generated by inputting the selected images from previous section and the designated prompt into each model. All interactions were conducted via the default web interface settings of ChatGPT Plus and Gemini Advanced, without modification. Each attempt was performed in a new chat session to clear the history and avoid any contextual influence from prior exchanges. The same prompt and settings were employed across all models to maintain methodological consistency and reproducibility.

3.3. Text evaluation

We employed several methods to evaluate the text generated by the LLMs. Initially, the length of the text generated by each model was measured to identify and compare their quantitative features. The qualitative analysis was conducted with the evaluation criteria of text meaning and text accuracy. The text generated by each model was divided into individual sentences and organized in a text analysis table. This table was used to categorize the semantic content and assess the correctness of each sentence. The meaning of the text was evaluated to assess the semantic diversity present in each text response generated by the LLMs. Text accuracy was included as an evaluation criterion to assess whether each semantic component provided a correct description. As there was no existing dataset of official descriptions of each image to use as a baseline, we relied on qualitative evaluations to classify the meaning of the sentences examine their correctness. Two researchers participating in qualitative evaluations were not blinded to the identity of the models or the input images during evaluation and independently conducted an initial assessment of each evaluation. After the independent evaluations, a joint review was performed in which disagreements and uncertainties were resolved through discussion. The researchers discussed the establishment of criteria that could be applied consistently to all model outputs, and resolved disagreements according to these criteria.

3.3.1. Text length

To measure the length of the text generated by each model, we counted the number of words in the text produced by each LLM. The word counts for each of the 23 images were obtained, and the text response for each image was divided into sentences, with the sentences numbered to determine the total number of sentences. Subsequently, the mean and standard deviation of the word and sentence counts were calculated for each model. This process provides insight into the average value and distribution of the length of each text response generated by the LLMs. Pairwise comparisons of mean length of response among the three models were conducted using Welch's two-sample t-tests (two-tailed). Statistical significance was determined using a p-value lower than 0.05.

3.3.2. Text meaning

We conducted a sentence-by-sentence analysis of the text to determine the meaning generated by each model without additional context, based solely on the image provided. To conduct the systematic analysis, text analysis tables were created using Google Spreadsheets. A distinct table was constructed for each of the three LLMs, with sentences from the text responses of all the images entered into each column. Two researchers, majoring in physics and computer science during their undergraduate studies, independently highlighted meaningful sentences in each model's text analysis table. In cases of disagreement, only sentences that were mutually agreed upon through further discussion were considered meaningful.

> *Evaluation criteria:* A sentence was considered meaningful if it provided a description or information relevant to the image. Sentences were excluded if they were not self-contained, did not provide information related to the image, or requested additional information from the user. For example, a statement that describes what is seen in an image, such as 'There is a prominent yellow streak near 0.0 seconds, centered around the 128 Hz frequency.', or conveys information about an image, such as 'This type of image is commonly used in fields like seismology, astrophysics, or any other science that analyzes waveforms over time.', was considered a meaningful statement regardless of whether the statement is true or false. However, sentences that are merely used to simply point to later content and do not convey information on their own, such as 'Here's a detailed breakdown of the image', or are intended to interact with the user, such as 'Please let me know if you have any other questions.', were excluded from meaningful sentences.

Subsequently, we calculated the proportion of meaningful sentences relative to the total number of sentences generated by each model. The semantic content of the highlighted sentences was recorded in a table and grouped into similar semantic categories. If a sentence conveyed multiple meanings, it was assessed as a separate semantic unit, which allowed for duplicate categorization. Thus a sentence containing two different semantic elements was treated as two distinct sentences, each with a different meaning category. For example, the sentence 'The image is a time-frequency representation, also known as a spectrogram, likely depicting data from the Hanford Observatory during its O3 observing run.' was considered two sentences belong to two categories 'a. Recognition and explanation of spectrogram' and 'c. Data source explanations' respectively, because it simultaneously describes both the explanation of the spectrogram itself and the source of that data in the same sentence. For each model, the number of sentences in each category was counted and the proportion of total sentences per category was calculated. This process allowed us to identify the types of semantic content generated by the LLMs in response to the input images.

3.3.3. Text accuracy

To determine the accuracy of the models' descriptions, we analyzed the correct sentences from the meaningful sentences produced by each model. For each semantic unit of text generated by each LLM, we assessed whether the sentence provided an accurate description of that semantic unit. Two researchers conducted the initial assessment independently, marking each sentence as 'YES' or 'NO' in a table to indicate its correctness. During the initial independent assessment, each researcher marked 'NS' for statements with uncertain judgments. After the independent evaluations, the researchers discussed in person to establish criteria for correct sentences and conducted a joint review to ensure that these criteria were applied consistently across all evaluation results. During this process, the two researchers reached a consensus, and any discrepancies or uncertain judgments were resolved as either 'YES' or 'NO.'

> *Evaluation criteria:* A correct sentence was defined as one that is consistent with the given image, if the sentence contains a description of the image itself, or as one that is consistent with the facts, if it contains an explanation or inference that extends beyond the intrinsic image features. In the case of quantitative descriptions of axis values, only those whose scale ranges exactly matched with the figure were considered correct. For signal source inferences, only those that actually referred to known sources, such as transient noise artifacts or gravitational wave signals, corresponding to the input image were considered correct sentences.

Two researchers were raters of each response, and after independent assessment and discussion, a consensus was reached to determine the correctness of each statement. For each model and semantic category, the proportion of correct sentences to the total number of meaningful sentences was calculated. This analysis identified the number of correct sentences generated by each model and revealed variations in accuracy across semantic categories. Inter-rater agreement was calculated for the initial assessment, using Cohen's κ with 95% confidence intervals (CIs).

4. Results

This section presents an analysis of studies that have attempted to employ LLMs to generate descriptive text from image data in the field of gravitational wave astronomy. We investigated whether multimodal LLMs are capable of generating text describing spectrogram images from a gravitational wave detector

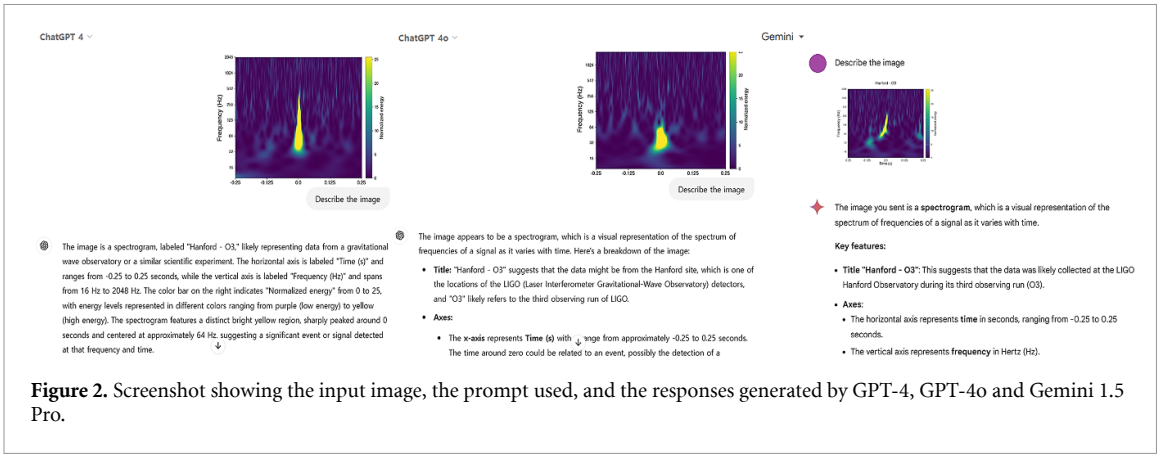


Figure 2. Screenshot showing the input image, the prompt used, and the responses generated by GPT-4, GPT-4o and Gemini 1.5 Pro.

without additional context and examined the quantitative and qualitative characteristics of the generated text.

4.1. Image collection

We downloaded the CSV file from the —'Gravity Spy Machine Learning Classification of LIGO Glitches from Observing Runs O1, O2, O3a, and O3b'— which is available on Zenodo, a publicly accessible repository for research data. Glitches are transient, non-Gaussian noises captured by gravitational wave detectors and can have a range of environmental and instrumental origins. The aforementioned file contains advanced LIGO data from the first three observing runs [45]. The Omega scan images contained within the file were selected from the dataset. These images are glitch spectrogram images captured by the Omicron transient search algorithm [46] during O1–O3 in the LIGO detector. These images were represented at four time intervals: 0.5, 1, 2, and 4 s. The Omicron algorithm identifies the potential noise transients by triggering excess power in the data stream. The Omicron software relies on the Q-transform, which is an analog of the short Fourier transform. Times of excess power are identified through the Q-transform, and the Omicron algorithm annotates each identified transient with characteristics such as event time, peak frequency, central frequency, and signal-to-noise ratio. The glitch morphology of the trigger can be visualized in a time—frequency spectrogram image.

The generation and analysis of descriptions of these spectrogram images is worthwhile because they can help classify and mitigate glitches in gravitational wave data. Understanding the characteristics of diverse glitches contributes to detector characterization and removal of glitches that interfere with the detection of real gravitational wave signals. Therefore the classification of glitches based on their morphological characteristics is an important task in gravitational wave data quality analysis. As mentioned in the Related works section, previous studies have relied on human scientists or volunteers to write text describing glitch morphology. This descriptive text can be used as a basis for categorizing or defining new classes of glitches when previously unseen morphologies are discovered. In accordance with a recently published update on glitch class classification [47], we collected representative images from each of the 23 glitch classes. We selected images that were morphologically similar to the sample images of each glitch class presented in this study [47], which clearly illustrated the characteristics described for each glitch class. The selected images were saved in PNG format, each with a resolution of 800×600 pixels.

4.2. Text generation

We generated text from the image dataset collected in the previous section by directly connecting to the ChatGPT and Gemini web interfaces and inputting prompts and images. For each image, the initial response generated by inputting the PNG image along with the prompt 'Describe the image' was used for analysis. Figure 2 illustrates how each LLM chatbot generated a description of the input image. The text generation results demonstrate that LLMs are capable of recognizing the image as a spectrogram related to gravitational waves. They provided descriptions relevant to the signal and interpreted them effectively without the need for additional context.

4.3. Text evaluation

4.3.1. Text length

Table 1 presents the averages and standard deviations of the word and sentence counts for the responses generated by each LLM model for the 23 images. For the 23 responses, GPT-4 had an average word

Table 1. Average word and sentence counts with standard deviations in the responses generated for each image.

	GPT-4	GPT-4o	Gemini 1.5 Pro
Word counts	159.43 ± 31.92	230.26 ± 41.50	236.83 ± 43.72
Sentence counts	7.13 ± 1.66	11.65 ± 2.27	13.22 ± 2.47

Table 2. Proportion of meaningful sentences to total sentences in LLM-generated responses (meaningful/total).

	GPT-4	GPT-4o	Gemini 1.5 Pro
Meaningful sentences	96.95% (159/164)	94.03% (252/268)	96.71% (294/304)

count of 159.43, GPT-4o had 230.26, and Gemini 1.5 Pro had 236.83. The average number of sentences per response was 7.13 for GPT-4, 11.65 for GPT-4o, and 13.22 for Gemini 1.5 Pro. Both the average word and sentence counts were the highest for Gemini 1.5 Pro, followed by GPT-4o and GPT-4, respectively. The standard deviation followed the same pattern, with larger variations observed for Gemini 1.5 Pro, followed by GPT-4o, and GPT-4. GPT-4 generated shorter responses than the other two models. Pairwise comparisons showed significantly lower average word and sentence counts for GPT-4 than for GPT-4o and Gemini 1.5 Pro (both $p < 0.001$, two-tailed Welch t-tests). In contrast, the average word count did not differ significantly between GPT-4o and Gemini 1.5 Pro ($p = 0.60$). Across all models, the responses contained an overall average of 208.84 words and 10.67 sentences per each image.

4.3.2. Text meaning

We identified sentences generated by LLMs that were considered meaningful and included descriptions of the given image or information relevant to the image. These sentences were then highlighted among all the sentences in the text analysis table. Table 2 shows the number of highlighted sentences generated by each LLM, as well as their proportion relative to the total number of sentences produced. GPT-4 had the highest proportion of highlighted sentences (96.95%), followed closely by Gemini 1.5 Pro (96.71%), and GPT-4o (94.03%).

We documented the meaning of each highlighted sentence, divided it into semantic units, and categorized them based on semantic similarity. Each meaningful highlighted sentences belonged to at least one semantic category. Tables 3–5 illustrate the distribution of the categorized sentences generated by GPT-4, GPT-4o, and Gemini 1.5 Pro, respectively, for each image. Table 6 summarizes the results for all three models and semantic categories. The number of sentences presented in the table is based on semantic units, which allows a sentence to have multiple meanings and be included in two or more categories simultaneously. When analyzing the text on a semantic basis, GPT-4 generated nine distinct meanings, GPT-4o generated ten meanings, and Gemini 1.5 Pro generated twelve meanings. LLMs generated the following semantic categories:

a. Recognition and explanation of spectrograms: Recognizing that an image is a spectrogram, and providing a brief explanation of what a spectrogram is.

b. Title descriptions: Recognizing and describing the title at the top of spectrograms.

c. Data source explanations: Comments about the detector that collected the signal data.

d. Axis descriptions: Explanations of the horizontal and vertical axes of the spectrogram mean, their labels, and value ranges.

e. Color descriptions: Descriptions of the color axis adjacent to the spectrogram, including labels, value ranges, and what each color represents.

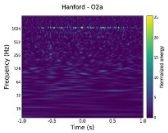
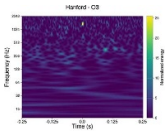
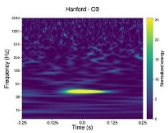
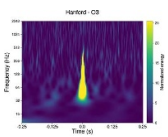
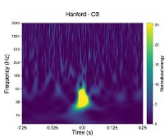
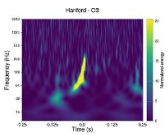
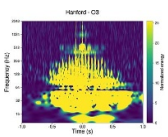
f. Explanations of the signal in the spectrogram: Descriptions of the bright signal in a given spectrogram image or an interpretation of its meaning.

g. Signal source inferences: Speculations about what caused the signal seen in the spectrogram.

h. Spectrogram background descriptions: Descriptions or interpretations of the background features that are not the main signals in the spectrogram described in category f.

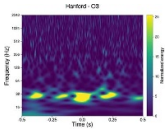
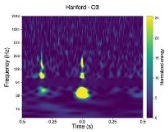
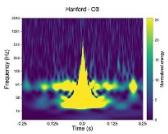
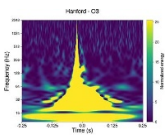
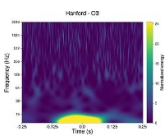
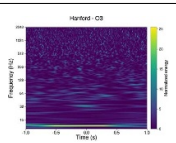
i. Applications of spectrograms: Explanations of where the spectrogram representation is used or specific examples of how it has been utilized.

Table 3. Number of meaningful sentences for each image generated by GPT-4.

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
1	1080 Lines		1	1	1	2	1	1	0	1	1	0	0	0	9
2	1400 Ripples		1	1	1	2	1	1	0	0	1	0	0	0	8
3	Air com-pressor		1	1	1	1	1	1	1	1	0	0	0	0	8
4	Blip		1	1	1	1	1	1	0	0	0	0	0	0	6
5	Blip low-fre-quency		1	1	1	2	1	2	1	1	0	0	0	0	10
6	Chirp		1	1	1	2	1	2	0	0	0	0	0	0	8
7	Extremely Loud		1	1	1	2	2	1	0	0	1	0	0	0	9

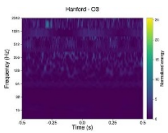
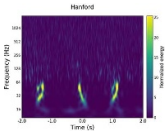
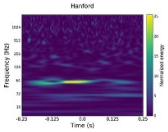
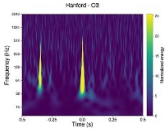
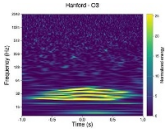
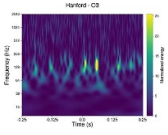
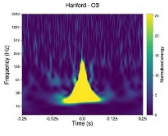
(Continued.)

Table 3. (Continued.)

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
8	Fast Scattering		1	1	1	2	2	0	0	0	1	0	0	0	8
9	Helix		1	1	1	2	2	1	1	0	1	0	0	0	10
10	Koi Fish		2	1	1	2	1	3	0	0	0	0	0	0	10
11	Light Modulation		1	1	1	2	2	1	0	0	1	0	0	0	9
12	Low-Frequency Burst		1	1	1	2	1	1	0	0	0	0	0	0	7
13	Low-Frequency Lines		1	1	1	4	2	2	0	0	2	0	0	0	13

(Continued.)

Table 3. (Continued.)

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
14	No Glitch		1	1	1	2	1	0	0	0	1	0	0	0	7
15	Paired Doves		1	1	0	1	1	2	0	0	1	0	0	0	7
16	Power Line		1	1	0	1	1	1	0	1	1	0	0	0	7
17	Repeating Blip		1	1	1	2	1	1	1	0	0	0	0	0	8
18	Scattered Light		1	1	0	2	1	1	0	0	1	0	0	0	7
19	Scratchy		1	1	1	2	1	2	1	0	0	0	0	0	9
20	Tomte		1	0	1	2	1	3	0	0	0	0	0	0	8

(Continued.)

Table 3. (Continued.)

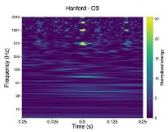
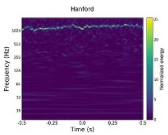
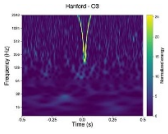
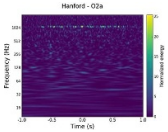
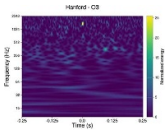
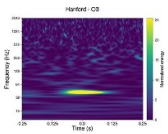
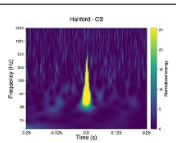
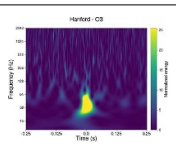
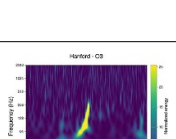
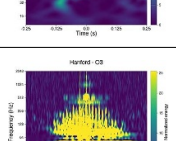
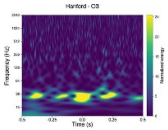
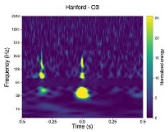
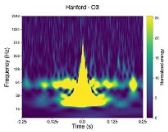
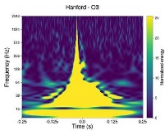
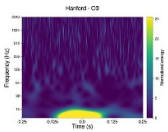
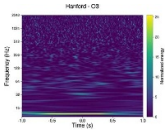
No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
21	Violin Mode		1	1	1	2	2	1	1	0	0	0	0	0	9
22	Wandering Line		1	1	0	1	1	1	0	1	0	0	0	0	6
23	Whistle		1	1	1	2	1	2	0	0	0	0	0	0	8
		Total	24	22	19	43	29	31	6	5	12	0	0	0	191

Table 4. Number of meaningful sentences for each image generated by GPT-4o.

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
1	1080 Lines		1	1	2	3	2	3	0	0	1	0	0	0	13
2	1400 Ripples		1	1	1	2	1	3	0	0	0	0	0	0	9
3	Air com-pressor		1	1	1	2	2	2	1	1	1	0	0	0	12
4	Blip		1	1	1	2	2	1	2	0	0	0	0	0	10
5	Blip low-fre-quency		1	1	1	4	2	2	0	0	1	0	0	0	12
6	Chirp		1	1	1	2	2	2	1	1	0	0	0	0	11
7	Extremely Loud		1	1	2	2	2	1	2	0	0	0	0	0	11

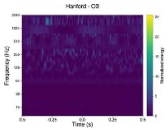
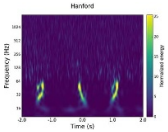
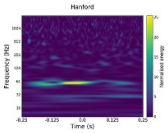
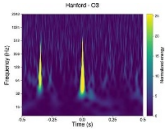
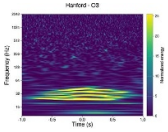
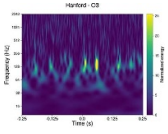
(Continued.)

Table 4. (Continued.)

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
8	Fast Scattering		1	1	1	2	3	2	2	0	0	0	0	0	12
9	Helix		1	1	1	2	2	3	1	1	0	0	0	0	12
10	Koi Fish		1	1	1	2	2	3	0	0	0	1	0	0	11
11	Light Modulation		1	1	1	3	1	2	1	0	0	0	0	0	10
12	Low-Frequency Burst		1	1	2	2	2	3	0	0	1	0	0	0	12
13	Low-Frequency Lines		1	1	1	2	2	4	0	0	1	0	0	0	12

(Continued.)

Table 4. (Continued.)

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
14	No Glitch		1	1	1	4	3	3	0	0	1	0	0	0	14
15	Paired Doves		1	1	1	2	2	3	0	0	0	1	0	0	11
16	Power Line		1	1	1	2	2	4	0	0	1	0	0	0	12
17	Repeating Blip		1	1	1	3	3	2	0	0	1	0	0	0	12
18	Scattered Light		1	1	1	4	2	5	0	0	1	0	0	0	15
19	Scratchy		1	1	2	3	3	2	1	0	2	0	0	0	15

(Continued.)

Table 4. (Continued.)

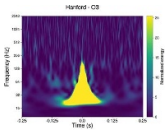
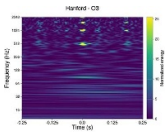
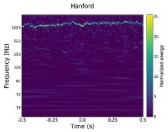
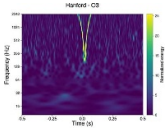
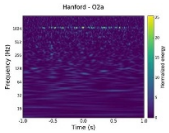
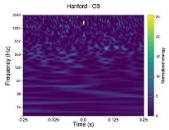
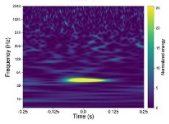
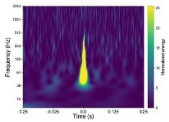
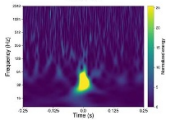
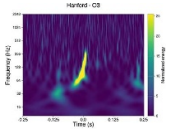
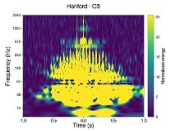
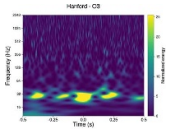
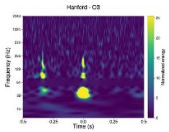
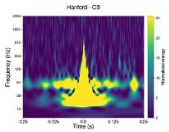
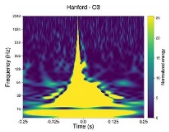
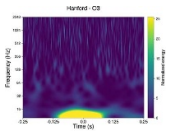
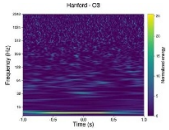
No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of interpretation	l. Need forfurther analysis	Total
20	Tomte		1	1	1	2	2	1	1	0	0	0	0	0	9
21	Violin Mode		1	1	0	2	2	2	0	0	1	0	0	0	9
22	Wandering Line		1	1	1	4	2	2	0	1	0	1	0	0	13
23	Whistle		1	1	1	5	4	4	0	1	1	0	0	0	18
		Total	23	23	26	61	50	59	12	5	13	3	0	0	275

Table 5. Number of meaningful sentences for each image generated by Gemini 1.5 Pro.

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of inter-pretation	l. Need forfurther analysis	Total
1	1080 Lines		1	1	0	4	2	2	0	0	1	1	0	0	12
2	1400 Ripples		1	1	2	2	1	2	1	1	0	1	0	1	13
3	Air com-pressor		1	1	1	2	2	1	3	1	0	0	1	3	16
4	Blip		1	0	1	1	1	2	1	0	0	1	1	3	12
5	Blip low-fre-quency		1	0	1	2	1	2	1	1	0	0	1	2	12
6	Chirp		1	1	2	2	2	2	2	0	0	0	0	0	12
7	Extremely loud		1	1	1	2	2	3	2	0	0	0	1	0	13

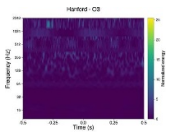
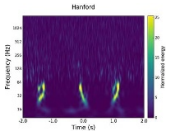
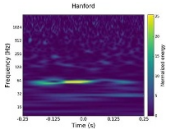
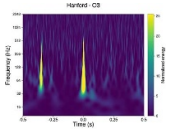
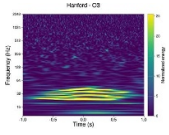
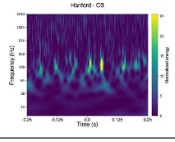
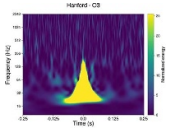
(Continued.)

Table 5. (Continued.)

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of inter-pretation	l. Need forfurther analysis	Total
8	Fast Scattering		1	0	1	3	2	3	0	1	1	0	0	0	12
9	Helix		1	0	1	2	2	2	1	1	0	0	1	2	13
10	Koi Fish		1	0	1	4	2	3	3	0	0	0	1	0	15
11	Light Modulation		1	0	2	3	2	3	3	1	0	0	1	0	16
12	Low-Frequency Burst		1	0	2	4	2	3	1	1	1	0	0	0	15
13	Low-Frequency Lines		1	0	2	4	2	2	2	1	0	0	0	2	16

(Continued.)

Table 5. (Continued.)

No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of inter-pretation	l. Need forfurther analysis	Total
14	No Glitch		1	1	1	2	1	3	4	0	0	0	0	5	18
15	Paired Doves		1	1	1	2	2	3	1	1	0	0	0	1	13
16	Power Line		1	1	2	2	1	1	1	1	0	0	0	0	10
17	Repeating Blip		1	0	1	2	2	2	1	1	1	0	0	1	12
18	Scattered Light		1	1	1	2	3	4	1	1	0	0	1	1	16
19	Scratchy		1	1	2	4	2	3	1	0	0	0	1	3	18
20	Tomte		1	0	1	2	1	3	1	0	0	0	1	1	11

(Continued.)

Table 5. (Continued.)

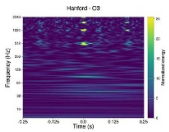
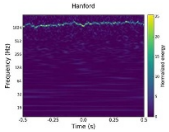
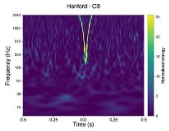
No.	Glitch class name	Image	a. Recognition and explanation of spectrogram	b. Title descrip-tions	c. Data source explanations	d. Axis descrip-tions	e. Color descrip-tions	f. Explanations of the signal in the spectrogram	g. Signal source inferences	h. Spectrogram background descriptions	i. Applications of spectrogram	j. Overall interpretation and summary	k. Limitations of inter-pretation	l. Need forfurther analysis	Total
21	Violin Mode		1	1	1	2	1	2	3	0	0	1	1	0	13
22	Wandering Line		1	0	1	2	1	1	2	1	0	0	1	1	11
23	Whistle		2	1	1	4	1	2	3	0	0	0	0	1	15
		Total	24	12	29	59	38	54	38	13	4	4	12	27	314

Table 6. Number of meaningful sentences for each semantic category and their proportion relative to the total number of meaningful sentences.

Semantic categories of text	GPT-4	GPT-4o	Gemini 1.5 Pro
a. Recognition and explanation of spectrogram	24 (12.57%)	23 (8.46%)	24 (7.64%)
b. Title descriptions	22 (11.52%)	23 (8.46%)	12 (3.82%)
c. Data source explanations	19 (9.95%)	26 (9.56%)	29 (9.24%)
d. Axis descriptions	43 (22.51%)	61 (22.43%)	59 (18.79%)
e. Color descriptions	29 (15.18%)	50 (18.38%)	38 (12.10%)
f. Explanations of the signal in the spectrogram	31 (16.23%)	59 (21.69%)	54 (17.20%)
g. Signal source inferences	6 (3.14%)	12 (4.41%)	38 (12.10%)
h. Spectrogram background descriptions	5 (2.62%)	5 (1.84%)	13 (4.14%)
i. Applications of spectrogram	12 (6.28%)	13 (4.78%)	4 (1.27%)
j. Overall interpretation and summary	—	3 (1.10%)	4 (1.27%)
k. Limitations of interpretation	—	—	12 (3.82%)
l. Need for further analysis	—	—	27 (8.60%)
Total	191	275	314

Table 7. Initial judgment agreement rate excluding sentences requiring consensus, and Cohen's κ coefficient.

	GPT-4	GPT-4o	Gemini 1.5 Pro
Initial judgment agreement	65.45% (125/191)	79.27% (218/275)	82.17% (258/314)
κ	0.12	0.30	0.49

j. Overall interpretation and summary: A brief interpretation and summary of the spectrogram data.

k. Limitations of interpretation: The difficulty in identifying a signal without additional context.

l. Need for further analysis: Mentions of the need for further analysis to gain deeper insights into, or additional interpretations of the signal, and specification of the information required for such analysis.

For all models, the most prevalent category of sentences in the generated text was 'd. Axis descriptions.' This can be attributed to the intrinsic nature of the spectrograms, which cause the LLMs to describe the horizontal and vertical axes as separate sentences, resulting in a relatively high number of such sentences. Following axis descriptions, the second most frequent category was 'f. Explanations of the signal in the spectrogram.' In comparison to the other two models, Gemini 1.5 Pro not only produced the highest average word and sentence counts but also generated the most semantically diverse text, encompassing 12 distinct semantic categories.

4.3.3. Text accuracy

We analyzed whether the meaningful sentences highlighted in Text Meaning correctly described their intended meanings. Each meaningful sentence was evaluated for accuracy as described in Text Accuracy in method section. If a sentence had multiple meanings, it was evaluated as multiple separate sentences for each meaning category. In cases where the two researchers" initial assessments were inconsistent or uncertain, a consensus was reached after discussion based on agreed-upon criteria. The number of sentences requiring consensus was 66, 57, and 56 for GPT-4, GPT-4o, and Gemini 1.5 Pro, respectively.

Table 7 presents the initial judgment agreement rate and inter-rater agreement for text accuracy evaluation. The initial judgment agreement rate refers to the proportion of the sentences for which both researchers' initial judgments were consistent relative to the total number of meaningful sentences evaluated. The number of sentences was calculated by subtracting cases requiring consensus (where the two researchers disagreed or at least one marked the judgment as uncertain) from the total meaningful sentences. For GPT-4, 125 of 191 sentences (65.45%) received consistent initial judgments without requiring consensus, 218 of 275 (79.27%) for GPT-4o and 258 of 314 (82.17%) for Gemini 1.5 Pro. The inter-rater agreement between the two researchers was measured using Cohen's κ coefficient, calculated based on the initial assessments of agreement and disagreement only. Uncertain judgments were excluded. Cohen's κ for inter-rater agreement in the text accuracy evaluation was 0.12 (95% CI: $-0.12 - 0.35$) for GPT-4, 0.30 (95% CI: $0.05 - 0.55$) for GPT-4o, and 0.49 (95% CI: $0.29 - 0.69$) for Gemini 1.5 Pro. We used the following level definitions for κ values: 0.20 or less indicates slight agreement, $0.21 - 0.40$ indicates fair agreement, $0.41 - 0.60$ indicates moderate agreement, $0.61 - 0.80$ indicates substantial agreement,

Table 8. Proportion of correct sentences relative to all meaningful sentences in each semantic category (correct/total).

Semantic categories of text	GPT-4	GPT-4o	Gemini 1.5 Pro
a. Recognition and explanation of spectrogram	100% (24/24)	100% (23/23)	100% (24/24)
b. Title descriptions	100% (22/22)	100% (23/23)	100% (12/12)
c. Data source explanations	100% (19/19)	100% (26/26)	100% (29/29)
d. Axis descriptions	48.84% (21/43)	62.3% (38/61)	71.19% (42/59)
e. Color descriptions	44.83% (13/29)	90% (45/50)	97.37% (37/38)
f. Explanations of the signal in the spectrogram	51.61% (16/31)	83.05% (49/59)	72.22% (39/54)
g. Signal source inferences	16.67% (1/6)	8.33% (1/12)	50% (19/38)
h. Spectrogram background descriptions	100% (5/5)	100% (5/5)	92.31% (12/13)
i. Applications of spectrogram	100% (12/12)	100% (13/13)	100% (4/4)
j. Overall interpretation and summary	—	100% (3/3)	100% (4/4)
k. Limitations of interpretation	—	—	100% (12/12)
l. Need for further analysis	—	—	100% (27/27)
Total	69.63% (133/191)	82.18% (226/275)	83.12% (261/314)

and $0.81 - 1.00$ indicates almost perfect agreement [48]. According to this criterion, the text accuracy evaluation revealed slight agreement between the two raters for GPT-4, fair agreement for GPT-4o, and moderate agreement for Gemini 1.5 Pro.

Table 8 presents the number and percentage of correct sentences relative to meaningful sentences generated by each LLM. The analysis demonstrated that Gemini 1.5 Pro correctly explained 83.12% of the total meaning, GPT-4o explained 82.18%, and GPT-4 explained 69.63% of the meaning. Although the ratio varied across the categories, all three models consistently demonstrated high accuracy (100%) in recognizing and describing spectrograms, describing titles, explaining data sources, and providing general information regarding spectrogram applications. However, the lowest percentage of correct sentences for all models was observed in signal source inferences, with GPT-4 and GPT-4o achieving only 16.67% and 8.33% accuracy, respectively, and Gemini 1.5 Pro performing relatively better at 50%. The accuracy was also relatively poor in the axis descriptions, color descriptions, and explanations of the signal in the spectrogram, which required more attention to detail in the spectrogram image. These results indicate that the models were able to accurately describe general information about the spectrogram itself, titles, and data sources which could be easily inferred from the spectrogram title. In contrast, the number of correct sentences decreased when describing the axis values, color axis values, and signals in the spectrogram and when explaining the detailed information or unique characteristics of the images.

5. Discussion

In this study, we analyzed the text responses of multimodal LLMs to gravitational wave spectrogram images. In this section, we reflect on our findings and discuss the limitations of our study as well as potential directions for future research.

5.1. Findings and discussion

5.1.1. Variation in response content across models

First, among the three LLMs, Gemini 1.5 Pro generated text with the most diverse meanings (12 semantic categories, the highest among the models) and was the only model to mention the limitations of interpretation of the given information, while also suggesting the need for additional information. Unlike the GPT models, which provide general explanations without addressing interpretive limitations, Gemini 1.5 Pro explicitly noted constraints, such as, 'Without additional context, it is difficult to say definitely what caused this signal' or 'I cannot definitively confirm the nature of the signal without more context.' It then clearly outlined the possible interpretations and provided the information required for a detailed interpretation. For instance, it stated, 'A more detailed interpretation would require further information about the specific event and the detector's characteristics.'

In other studies comparing the responses of GPT-4 and Gemini to the same image, we found that Gemini's responses were qualitatively more concise than those generated by GPT-4 or other models. For example, Reith *et al* [14] found that Gemini's responses were less verbose than those of GPT-4 and Claude when generating radiological image diagnoses. Another study of medical images conducted by Khan *et al* [34] concluded that Gemini provided concise and confident interpretations of medical images, whereas GPT-4 V tended to generate more general information about x-ray imaging and structure. Our

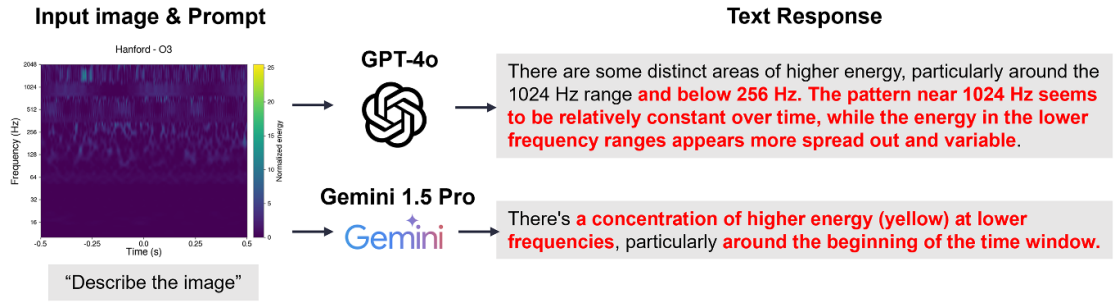


Figure 3. Examples of incorrect descriptions generated by multimodal LLMs. The text with red highlights are the incorrect descriptions in the image, where there is no visible concentration of high energy in the spectrogram image. Permission for the inclusion of the Chat GPT and Gemini logos have not been sought. The inclusion of the Chat GPT and Gemini logos is not intended to indicate any endorsement by or collaboration of Chat GPT or Gemini with this publication or any other IOP Publishing material.

findings align with these observations. These findings suggest that the differences in the number of semantic categories may be attributed to the qualitative response characteristics exhibited by each model. Unlike the GPT models, which tend to provide generic statements, Gemini explicitly acknowledges interpretive limitations, highlights the need for additional information, and provides targeted interpretations, similar to the qualitative evaluation results observed in other studies.

Further discussion is possible regarding the relationship between these response characteristics and other features of models' answers. In quantitative evaluation, Gemini 1.5 Pro produced the longest text and the highest number of meaningful sentences, as well as the highest accuracy and inter-rater agreement. In contrast, GPT-4 produced the shortest text and the lowest text accuracy. In qualitative evaluation, GPT-4's responses frequently exhibited a tendency to provide generic content without adequately reflecting the specific characteristics of the input image. This tendency may, in part, contribute to the model's inferior performance in accurately describing spectrograms (see table 7). Taken together, these findings indicate that generating longer responses with LLMs does not necessarily result in vagueness or additional errors. Instead, the quality of the response may appear to be more strongly influenced by the intrinsic characteristics of the model than by response length itself. For instance, GPT-4o and Gemini 1.5 Pro were released around the same period and showed no substantial difference in average word counts. However, Gemini's broader semantic coverage and clear response style appear to yield more meaningful descriptions than quantitative accuracy alone, which is likely reflected in its higher inter-rater agreement.

5.1.2. Generation of incorrect signal descriptions

Second, the LLMs generated descriptions of features that were not visible in the spectrogram image. For example, figure 3 illustrates the incorrect text responses generated by GPT-4o and Gemini 1.5 Pro in response to an image. In figure 3, the input image is a glitch named 'No Glitch', designated for spectrograms that have no apparent or obvious transient noise structure visible in the Omega scan. However, both multimodal LLMs produced signal depictions that were not visible in the spectrograms due to hallucinations.

Similar results have been found in other studies that attempted to use multimodal LLMs to interpret images in specific domains, similar to our study. Lee *et al* [40] found that Gemini Pro mistakenly recognized a student-created scientific model image as a scientific poster, and generated hallucinated information. Wang *et al* [15] employed GPT-4 V to generate interpretations of complex bioinformatics plots and found that it faced challenges in interpreting colors and conducting quantitative analyses. The generation of incorrect signal descriptions may stem from several factors. First, hallucination effects may occur when the model overgeneralizes from unrelated training examples and generates plausible but incorrect features. Chatbots tend to exhibit hallucinations when conducting tasks outside the scope of their training. Second, limitations in the training data, particularly the scarcity of domain-specific gravitational wave spectrograms, reduce the model's ability to map visual patterns to accurate terminology. This problem may stem from our experimental setting, which used a context-free prompt and provided no supplementary textual cues or additional information. Therefore, it is probable that the generation of descriptions of nonexistent signal features by LLMs is a hallucination, perhaps due to the task being outside of the training data [49]. It also seems that these concerns could be compensated for by creating chatbots that specialize in specific contexts with sufficient pre-training.

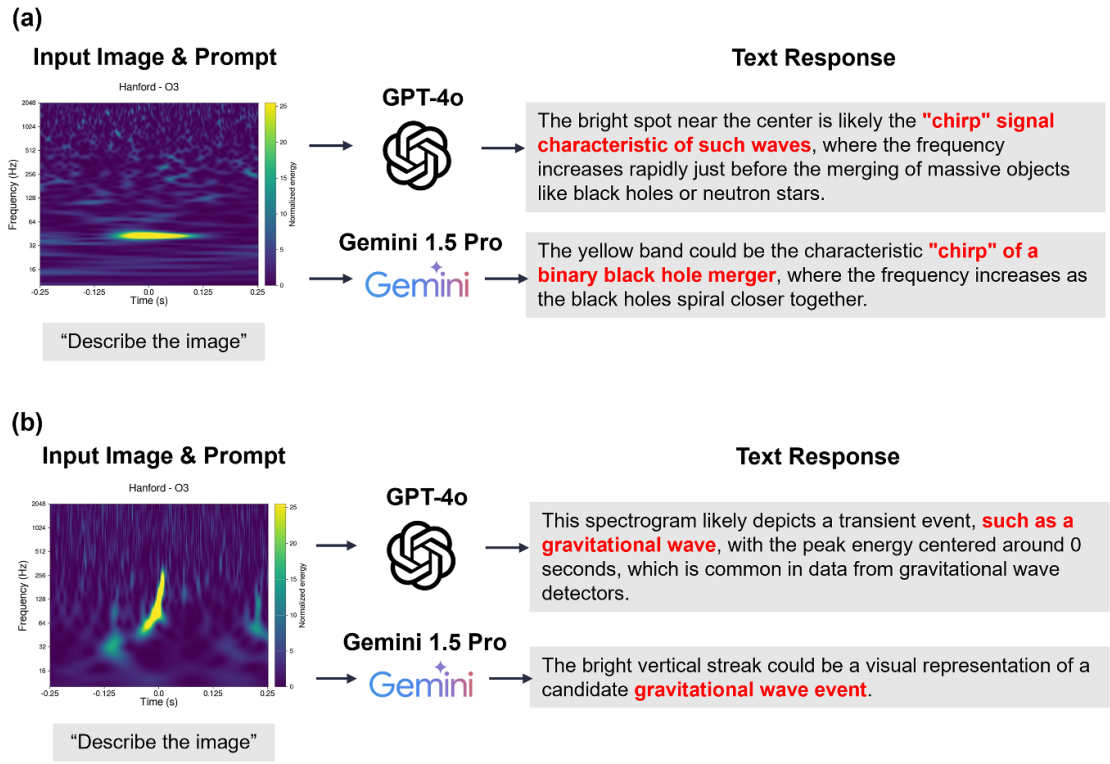


Figure 4. Examples of signal source inferences generated by GPT-4o and Gemini 1.5 Pro. Text highlighted in red indicates the possible source of the signal mentioned by the models. (a) Incorrect inference for image 3, a spectrogram of the 'Air Compressor' class glitch. (b) Correct inference for image 6, a 'Chirp' class signal corresponding to the spectrogram of the gravitational wave event. Permission for the inclusion of the Chat GPT and Gemini logos have not been sought. The inclusion of the Chat GPT and Gemini logos is not intended to indicate any endorsement by or collaboration of Chat GPT or Gemini with this publication or any other IOP Publishing material.

5.1.3. Difficulty in matching signal class and signal source

Third, although LLMs provided correct information for general explanations, they demonstrated notable deficiencies in accurately describing detailed features and inferring the source of the signal. These findings indicate that LLMs struggled to identify accurate physical behavior in the one-shot, zero-context setting of this study, where no additional information was provided beyond the image itself. LLMs often made errors in associating different class names with given signals and failed to provide meaningful interpretations of signal spectrograms. The models provided incorrect interpretations of glitch images that were not gravitational wave signals, such as 'This spectrogram could represent a gravitational wave signal detected at LIGO Hanford.' For instance, only image 6 among the 23 input images corresponded to a 'Chirp' class signal, which is believed to be characteristic of gravitational waves emitted by astronomical events. However, the LLMs incorrectly stated that the signals in Images 3, 7, 10, 11, 14, 20, and 23 also exhibited features of a chirp signal (see table 3 for images).

Figure 4 shows an example of this common misclassification. In figure 4(a), we present part of each model's answer to image 3. image 3 is a glitch spectrogram image corresponding to the 'Air Compressor' class. In the Hanford Detector, these kinds of transient noises were found to be related to air compressor motors, and were reduced following the replacement of the vibration isolators [47]. However, GPT-4o and Gemini 1.5 Pro incorrectly match this glitch class as having the characteristic of 'Chirp' and also mention the astronomical event, such as black holes merging, as a signal source. The spectrogram image that was actually classified as 'Chirp' is the image provided in figure 4, which was found to correspond to a real gravitational wave event spectrogram, GW200129_065458 [50]. Otherwise, LLMs often generated vague speculations like 'The characteristics suggest it could be a transient event, such as a gravitational wave signal from a compact binary coalescence or a glitch from the detector itself.' Such responses did not help readers distinguish whether the image depicted a gravitational wave signal or noise. Future research should further explore the limitations of this interpretation to determine how incorporating few-shot learning or contextualized prompt refinement might improve it.

The advent of multimodal LLMs has introduced a novel approach to generating image descriptions across various domains, providing interpretations that assist in decision-making. Nevertheless, recent

studies evaluating the ability of LLMs to interpret radiological and electrocardiographic images have concluded that their current use in medical image interpretation and diagnosis remains risky. Günay *et al* [36] concluded that the limited accuracy and consistency of GPT-4 and Gemini make their current use in clinical electrocardiography (ECG) interpretation risky. Similarly, Hindy *et al* [38] found that both models exhibit limitations in distinguishing certain bacterial images with an accuracy rate of 0% in specific instances. These findings indicate that although multimodal LLMs show promise, they are not yet ready for implementation in clinical practice. Our finding is similar to the challenges that multimodal LLMs have encountered in other domains, such as biological and medical image interpretation. In the absence of supplementary information or contextual data, LLMs often failed to interpret and inference the signal spectrogram adequately. This suggests that, under constrained conditions, the current capacity of multimodal LLMs is insufficient for scientific image interpretation. Further studies with richer prompts, domain-specific training, and broader evaluation settings are needed to fully assess their potential.

5.2. Limitations and future work

5.2.1. Prompt variation

Our study has several limitations. First, the evaluation was limited to a one-shot, zero-context setting without prompt engineering. The LLMs received only the input from the images and the minimal prompt, without supplementary information or contextual data beyond the embedded labels in the spectrogram. This design enabled the evaluation of performance under strictly controlled, baseline conditions but it does not reflect the authentic multimodal capabilities of the models. Providing more sophisticated prompts, such as 'Please act as a physicist' or 'Be as specific as possible,' with prior context information, potentially improves the quality and accuracy of the generated descriptions. An additional investigation will be conducted to ascertain how the inclusion of additional information or context affects the descriptive and interpretive performance of multimodal LLMs.

The follow-up study requires sophisticated prompts for scientific figure analysis and contextual information of the input images. The prompt engineering strategies will include role-based instructions and domain-specific requests accompanied by supplementary information. Since gravitational wave detectors are affected by a lot of information, such as detector information, environment, and known noise sources, the prior context included in the prompts should be carefully researched and selected. However, the present study did not incorporate prompt variation, nor did it consider any supplementary information beyond single images of each glitch class. Moreover, current study did not establish evaluation methodologies for comparing the quality of responses to different prompts provided by the same model. Future work will address these limitations by incorporating prompt variations, richer contextual information, and standardized evaluation methods, enabling the clear evaluation of the models' genuine multimodal reasoning.

5.2.2. Embedded text in spectrogram images

Second, we did not control for the presence of embedded text in the spectrogram image, such as axis label, color bar, or figure title. These elements may have been partially or fully recognized by the models and could have influenced their responses. Specifically, the spectrogram title included the name of the detector from which the spectrogram was taken and the observation period. For example, the title 'Hanford-O3' is expected to have provided a significant cue to the model's response. Thus, the descriptions may be influenced by the interpretation of the textual information rather than solely by visual pattern recognition.

Future work will explicitly assess the impact of embedded text elements within the images, such as axis labels, color bars, and figure titles, by comparing model responses to images with and without such information. This comparison will help determine the extent to which the models rely on embedded text for generating descriptions and interpretation for scientific figures. Additionally, while the focus of our paper was on spectrogram images of gravitational waves, it should be noted that spectrograms represent just one method of visualizing gravitational wave data. Other representations, such as time series or raw waveform signals, can be provided as complementary information or just the input image itself. In that case, we can confirm the multimodal LLM's figure interpretation capability in various types of gravitational wave visualization in extended context.

5.2.3. Evaluation metric

Last, we did not evaluate the LLM-generated image description text by comparing it with the reference text. To analyze the meaning and accuracy of the LLM-generated sentences, we relied solely on the judgment of researchers. The availability of expert-written reference text for each image would have enabled

a quantitative analysis of text quality using various metrics to measure text quality, such as bilingual evaluation understudy or metric for evaluation of translation with explicit ordering. In the context of future model evaluations, we will develop multimodal datasets that combine spectrogram images of each glitch class with expert-written descriptions. By comparing LLM-generated text with these references, it is possible to quantitatively assess the quality and accuracy of each model's output. This approach will reduce reliance on purely qualitative judgments, and provide a more objective performance evaluation framework for gravitational wave data analysis. Such datasets could also be used to train or fine-tune specialized models for automated glitch classification, enabling more robust and scalable analysis pipelines.

We deliberately chose not to present the confusion matrix in the results section for signal source inference in models, which stems from our experimental design. Since we only instructed the models to 'describe' the spectrogram image, the models often did not explicitly mention the signal source but merely listed possible sources of the signal. Future work aims to quantify the performance of LLMs in signal source inference by employing a confusion matrix to represent their accuracy. To do so, we must prompt the model to provide an unambiguous answer of either gravitational wave signal or glitch. Further research incorporating targeted prompts and exemplar responses, would address the methodological limitations of the present study by facilitating systematic evaluation.

Future research will extend to integrating LLM-generated spectrogram descriptions into gravitational wave detector characterization workflows, enabling automated report generation for transient event analysis. Traditionally, the characterization of newly emerging glitches during detector upgrades has largely depended on manual inspection by experts and contributions from citizen scientists. The findings of this study suggest the potential for multimodal LLMs to be integrated into the parts of this data quality analysis process. Enhancing their ability to describe and interpret spectrogram images could accelerate the characterization and classification of detector noise. Furthermore, if their performance of image-to-text conversion has qualified, their applications could be extended to the analysis of gravitational wave data visualized in formats other than spectrograms. This broader applicability underscores their potential to support diverse stages of gravitational wave data analysis pipelines, not only mere image descriptive tasks, but also downstream objectives such as anomaly detection or signal classification with multimodal data. Beyond gravitational wave astronomy, this approach could serve as a model for incorporating AI-driven image interpretation into other areas of scientific research, where rapid and reliable analysis of complex visualization data is critical for decision-making.

6. Conclusion

The primary objective of this study was to investigate the feasibility of using multimodal LLMs for generating descriptive text from gravitational wave spectrogram images. This study represents the novel application of LLMs in the domain of gravitational wave astronomy. In particular, the combination of image data analysis in gravitational wave astronomy with image-to-text conversion using multimodal LLMs suggests promising potential for the transition to multimodal data analysis. Twenty-three spectrogram images from gravitational wave detectors were provided to three models, GPT-4, GPT-4o, and Gemini 1.5 Pro, and the generated text responses were analyzed. The results of the text evaluation demonstrated that all three models are capable of describing spectrogram images without prior context. They correctly recognized the spectrogram in the image and its relevance to gravitational wave detectors. The content of the generated text also included explanations of the structure and components of the spectrogram (e.g. axes and titles), descriptions and interpretations of the signal presented in the spectrogram. Notably, Gemini 1.5 Pro produced the longest and most semantically diverse responses, clearly outlining potential interpretations while explicitly acknowledging their limitations. However, all three models occasionally exhibited hallucinations and provided descriptions that diverged from the actual signal content. Although the models were able to generate a general explanation of the spectrogram correctly, they often struggled to accurately describe the signal and derive meaningful source inferences without additional information.

These findings indicate that in the absence of further context, the image interpretation capabilities of multimodal LLMs are insufficient for direct application to gravitational wave spectrogram analysis. However, this conclusion is constrained by the results obtained in a one-shot setting with a single image, without prior context or few-shot examples. In this study, we did not employ prompt engineering or provide additional contextual information related to the images, which presents an area for future research. Moreover, the evaluation was conducted without reference texts, which limited the clarity of the performance measures. Future work includes a quantitative comparison between human-generated

and LLM-generated texts to develop a structured evaluation framework. These efforts could also support the creation of a multimodal dataset of gravitational wave spectrograms and enable the integration of multimodal LLMs into report generation tasks within gravitational wave data analysis pipelines.

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Data availability statement

The data that supports the findings of this study are included within the article (and any supplementary files).

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